

Plan of Attack

1 Assignment

Now that your group has identified an organization, a research question, and a dataset, it is time to plan your analysis. A good plan of attack forces you to think carefully about *how* you will use the data to answer your question *before* you start running analyses. This matters because it is easy to get lost once you start coding, chasing interesting-looking patterns that may have nothing to do with your original question.

Your plan of attack should have three sections, described below. For Sections 2 and 3, you must explain *why* you are making each decision. Specifically: how does this step help you answer your overarching research question? If you cannot articulate why a step is necessary, you probably do not need it.

Your analyses going forward are not entirely locked in to what you submit here. Your approach may evolve as you learn more from the data, but be careful that it is not post-hoc accidental *p*-hacking! But writing this plan is roughly 90% of the intellectual work. If your plan of attack is clear enough and sharp enough, AI can actually read it and execute the entire analysis for you in one shot. Now with AI, the hard part is the thinking, not the coding.

At the end of this document, I link to example solutions. For the **causal-inference analyses**, you may consider these a “cook book” that you can attempt to follow closely. My aim is to set you up for success with algorithms you can apply to general causal-inference contexts. **Descriptive analyses**, by contrast, are easier conceptually but so specific that my example solution will not exactly map onto your descriptive problem, should you choose one. You will have to think a bit more carefully about what descriptive analyses are useful to help you answer your specific question.

Please submit your writeup on Canvas.

2 Deliverables

Section 1: Describe the Raw Data

Load your dataset into Positron and describe what you are working with (you can use AI to do this). This section should include:

- a. The number of observations and variables.

- b. What each row represents (the unit of observation). State what uniquely identifies each observation (e.g., “each observation is uniquely identified by a game date and a home team”).
- c. The time period the raw data covers.
- d. Counts of unique cross-sectional units that are plausibly relevant to your analysis. For example, if your data is at the game level, how many unique teams, stadiums, and seasons does it span? If your data is at the listing level, how many unique cities, hosts, or neighborhoods? These counts help the reader (and you) understand the scope of variation you are working with.
- e. Summary statistics for your key variables (at minimum, the dependent variable and the main independent variables you plan to use). Don’t just conduct summary statistics on every variable, even ones you have no plans of using. **Also visualize the distribution of each dependent variable** (a histogram for continuous variables; a count or proportion table for binary or categorical variables)—a summary statistic can hide important features of the distribution, such as extreme outliers, bimodality, or class imbalance. Scatter/line plots (dependent variable against time, or against each main independent variable) are also useful for spotting trends and high-leverage points.
- f. Any immediate data quality issues you notice: % missing values, suspicious outliers, duplicate rows, etc. This part should be done by variable.

If you are using multiple datasets (e.g., merging two sources together), describe each one separately.

Section 2: Sample Construction Plan

This section is about the decisions you will make to go from the raw data to an analysis-ready dataset. The purpose of laying this out explicitly is to credibly signal that there is nothing fishy (*p*-hacky!) in the mapping from your raw data to your core analysis sample. For each decision, explain *why* you are making it and how it connects to your research question.

Address the following:

- a. **Sample window.** What time period will you use, and why? Are there structural breaks in your data (e.g., a policy change, COVID, a merger, a new product launch) that might make some periods incomparable to others? If so, how will you handle them? (Advice: try to avoid contaminated parts of the data in a principled way, e.g., drop control units that also experience major policy changes or big structural shifts like COVID.)
- b. **Other sample restrictions.** Are there observations you plan to include or exclude? For example, will you drop certain geographies, subpopulations, or edge cases? Every

restriction should be justified by the research question, not by convenience. (Also see point (g).)

- c. **Dependent variable integrity.** Your dependent variable is the outcome you are trying to explain or predict. Check whether it is measured reliably. Are there missing values? Implausible values? How will you handle these cases, and why? If your dependent variable is continuous, also check for outliers that might distort your results: a common and principled approach is trimming or winsorization at a commonly used threshold (e.g., 95th, 97.5th, or 99th percentile). (Hint: Look up winsorization if you are not familiar with it, or ask AI about it.) If your dependent variable is binary or categorical, outlier and winsorization concepts do not apply; instead check for coding errors (e.g., NAs miscoded as 0), severe class imbalance, or drift over time. In general, the best approach is to avoid altering the raw data unless one is forced to, but extreme outliers in a continuous dependent or independent variable could justify winsorization or trimming.
- d. **Independent variable integrity.** Your key independent variables are the factors you believe drive the outcome. Apply the same checks. If you need to construct a variable (e.g., computing a win percentage from raw scores), describe exactly how you will do it.
- e. **Aggregation (if applicable).** If your raw data is at a finer level than your analysis requires (e.g., individual transactions but you need daily totals, or minute-level readings but you need only week-year data), describe how you will aggregate and why that level of aggregation is the right one for your research question. (Hint: if you have incredibly granular data so that a key variable is often “0”, aggregation to a higher level of time can reduce the frequencies of zeroes, which are tough to visualize in the raw data.)
- f. **Data merging (if applicable).** If you are combining multiple data sources, describe how you will join them and what you expect to lose in the merge (unmatched rows, etc.).
- g. **Report all filters and transformations.** For each filter you apply to the raw data, report how many observations (or what percentage of the data) it removes, if any. This is essential for transparency. If a single filter drops a large share of your data, you should explain why that is expected and why it does not threaten the validity of your analysis. For transformations (e.g., log transforms, winsorization, constructing new variables), describe what you did and why, but you do not need to report observation counts.

Section 3: Analysis Plan

This section describes what you will actually do with your cleaned data to answer the research question. Again, for each analysis, explain *why* it helps answer the question.

Address the following:

- a. **Research design.** State clearly whether your question is causal or descriptive/predictive.
 - If your question is **causal** (e.g., “Does X cause Y?”), you need an identification strategy—most likely one of the four covered in the Question and Dataset assignment: difference-in-differences (DID), synthetic control, regression discontinuity (RDD), or A/B test. State which one you are using. What source of variation are you exploiting to separate cause from correlation? What assumptions does your strategy require, and are they plausible (some of this content will be covered in week 6)?
 - If your question is **descriptive or predictive** (e.g., “What predicts Y?” or “How does Y vary across groups?”), describe your approach. What comparisons, models, or summaries will you produce?
- b. **Key analyses.** List the specific analyses you plan to run. You should run as many analyses as you need to answer the question, and nothing more. For each one, describe:
 - (i) A high-level summary of what you are testing.
 - (ii) Specific instructions that are precise enough for an AI to execute for you.
 - (iii) How this analysis contributes to answering the research question.

In addition to all the other analyses, think especially carefully about what your “Figure 1” should be. In other words, if you could show the reader only one picture to convey your main finding, what would it be? Describe it in words, and plan out your other key figures as well. You should sketch them by hand or have AI make fake versions of what you expect they will look like. (For perspective, most strong analyses have about three main figures, and the best papers tell their entire story through Figure 1 alone. That is a high bar, and it is perfectly fine if you do not meet it. Use that as a rough guide for how many key analyses you need.)

- c. **(Extra credit) Robustness checks.** Think about the biggest potential sources of fragility in your analysis. What are the assumptions or choices that, if changed, might cause your conclusion to fall apart? For each source of fragility, describe a robustness check that directly tests whether your result survives (or doesn’t!). Only run checks that matter. Do not pad this section with tests you do not care about. For each check, explain the specific concern that motivated it. Some examples of common robustness

checks:

- *Alternative sample window*: Re-run the analysis dropping the first or last year of data. Concern: results might be driven by unusual conditions at the boundary of the sample.
- *Alternative outlier treatment*: Winsorize at the 95th percentile instead of the 99th, drop outliers entirely, or apply no winsorization at all. No winsorization is often the most credible specification, though it may not be justified a priori if your initial investigation revealed extreme outliers. Concern: a few extreme observations might be driving the result.
- *Alternative variable definition*: Use a different measure of the dependent or independent variable. Concern: the result might be an artifact of how you defined the variable.
- *Placebo test*: Run the same analysis at a time or place where you would *not* expect to find an effect. For example, test for an “effect” backdating the treatment one year before the actual treatment (and dropping all dates after the real treatment begins). If you find an effect, then your result is driven by pre-existing differences between control and treatment rather than by the treatment (which had not occurred in the data yet!). Analogously, in a regression discontinuity, test at an arbitrarily chosen threshold where no real discontinuity exists. If you find an “effect” where there should be none, your main result is suspect.
- *Parallel trends (for difference-in-differences)*: If your analysis compares a treated group to a control group before and after an intervention, show that the two groups were trending similarly in the pre-intervention period (i.e., take the average of treated units and average of control units, and make a line plot for each as we did in the Italian Judges Lab). Concern: if the groups were already diverging before the intervention, the estimated effect may reflect a pre-existing trend rather than the intervention itself.

3 Example Solutions

Fully worked example solutions are available for each of the five cases below. Pick the one whose method most closely matches your own project and use it as a template.

Project type	Example case	Download
Difference-in-differences (DID)	Marijuana legalization	Download PDF
Synthetic control	Food waste bans	Download PDF
Regression discontinuity (RDD)	San Diego Padres	Download PDF
A/B test	Hillstrom email marketing	Download PDF

Project type	Example case	Download
Descriptive / predictive	Airbnb San Diego	Download PDF